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Stat 201 Final Exam Term 252

Original exam date: 13 May 2026. Practice version: questions and choices shuffled.
Net time allowed: 120 minutes

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1. (Problem 9-40) The mean water temperature downstream from a power plant cooling tower discharge pipe should be no more than 38.11°F. Past experience has indicated that the standard deviation of temperature is 1.55°F. The water temperature is measured on 25 randomly chosen days, and the average temperature is found to be 37.4°F. What is the p-value for testing

$$H_0 : \mu \leq 38.11 \quad \text{vs.} \quad H_1 : \mu > 38.11$$

(a) 0.01101

(b) 0.67724

(c) 0.02202

(d) 0.32276

(e) 0.98899

2. (Problem 4-60) Cholesterol is a fatty substance that is an important part of the outer lining (membrane) of cells in the body of animals. The Food and Nutrition Institute of the Philippines found that the total cholesterol level for Filipino adults has a mean of 160 mg/dl and 84.13% of adults have a cholesterol level less than 180 mg/dl. Suppose that the total cholesterol level is normally distributed. An adult is at moderate risk if cholesterol level is more than one but less than two standard deviations above the mean. What proportion of the population is at moderate risk according to this criterion?

(a) 0.8413

(b) 0.0228

(c) 0.1587

(d) 0.1359

(e) 0.9772

3. (Problem 3-16) $f(x) = \left(\frac{8}{7}\right)\left(\frac{1}{2}\right)^x$, $x = 1, 2, 3$. Determine $P(X \leq 1 \text{ or } X > 1)$.

(a) 1.0

(b) 0.0

(c) 0.25

(d) 0.5

(e) 0.75

4. (Problem 11-1) An article in Concrete Research presented data on compressive strength (x) and intrinsic permeability (y) of various concrete mixes and cures. Summary quantities are $n = 13$, $\sum y_i = 490.13$, $\sum y_i^2 = 18706.8045$, $\sum x_i = 54.86$, $\sum x_i^2 = 257.1384$, and $\sum x_i y_i = 1995.477$. Also,

$$S_{XX} = 25.6292, \quad S_{YY} = 227.77243, \quad S_{XY} = -72.8716, \quad SSR = 207.19609, \quad SSE = 20.57634.$$

Assume that the two variables are related according to the simple linear regression model. The fitted regression line is $\hat{y} = 49.70105 - 2.84330\,x$. What is the standard error of the slope coefficient?

(a) 0.89602

(b) 0.27016

(c) 1.36769

(d) 0.24851

(e) 0.07299

5. (Suggested Problem 7-12) The amount of time that a customer spends waiting at an airport check-in counter is a random variable with mean 8.3 minutes and standard deviation 1.8 minutes. Suppose that a random sample of $n = 36$ customers is observed. Approximate the probability that the average time waiting in line for these customers is between 8.0 and 8.6 minutes.

(a) 0.84134

(b) 0.34134

(c) 0.68268

(d) 0.13236

(e) 0.15866

6. (Problem 6-6) Suppose that you add 5 to all of the observations in a sample. How does it change the sample standard deviation?

(a) The standard deviation is increased by 5.

(b) The standard deviation is decreased by 5.

(c) The standard deviation is increased by 25.

(d) The standard deviation is not changed.

(e) The standard deviation is increased by 2.5.

7. (Problem 9-56) Consider the computer output shown in the figure below.
At what level of significance can the null hypothesis be rejected?

```
 One Sample t-test

data:  X
t = 2.858, df = 15, p-value = 0.012
alternative hypothesis: true mean is not equal to 34
95 percent confidence interval:
34.324 36.224
sample estimates:
mean of X
35.274
```

(a) We can reject the null hypothesis only at significance level 0.05.

(b) We can reject the null hypothesis at significance levels greater than or equal to 0.012.

(c) We can reject the null hypothesis at significance levels less than or equal to 0.012.

(d) We can reject the null hypothesis only at significance level 0.012.

(e) We can reject the null hypothesis at significance levels less than or equal to 0.988.

8. (Problem 8-28) An Izod impact test was performed on 20 specimens of PVC pipe. The sample mean is 1.25 and the sample standard deviation is 0.25. Compute a 98% confidence interval on average Izod impact strength, and use it to test the hypothesis $H_0 : \mu = 1.0$. Which one of the following statements is correct?

(a) Do not Reject H_0, as 1.0 falls outside the confidence interval

(b) Reject H_0, as 1.0 falls inside the confidence interval

(c) Reject H_0, as 1.0 falls outside the confidence interval

(d) Using the computed confidence interval, we can not reach a valid decision

(e) Do not Reject H_0, as 1.0 falls inside the confidence interval

9. (Problem 12-21) Consider the R output shown in the figure below for a multiple linear regression model (15 observations, two predictors x_1 and x_2). Entries marked ?? are missing. What is the value of the standard deviation of the residuals (standard error of estimate)?

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)   76.1272    10.6624   ??       ??
x1             2.6673     0.7666   3.4793   ??
x2            -2.7246     0.9865   ??       ??

Residual standard error: ?? on 12 degrees of freedom
Multiple R-squared:  ??,    Adjusted R-squared:  0.9972
F-statistic: ?? on 2 and 12 DF,  p-value: ??

Analysis of Variance
                Df    Sum Sq   Mean Sq   F value   Pr(>F)
Regression       2  1366.288   683.144     ??       ??
Residual Error  ??     ??        ??
Total           14  1369.613
```

(a) 0.7666

(b) 0.5264

(c) 0.2771

(d) 0.4709

(e) 3.4793

10. (Problem 9-1) Which of the following hypotheses refer to a correctly stated hypothesis testing problem?

(a) $H_0 : \mu = 50, \quad H_1 : \mu \neq 50$

(b) $H_0 : p \neq 0.1, \quad H_1 : p = 0.1$

(c) $H_0 : \bar{x} = 50, \quad H_1 : \bar{x} \neq 50$

(d) $H_0 : s = 30, \quad H_1 : s > 30$

(e) $H_0 : \sigma > 10, \quad H_1 : \sigma = 10$

11. (Problem 9-20) A manufacturer is interested in the output voltage of a power supply used in a PC. Output voltage is assumed to be normally distributed, with standard deviation 0.24 volt, and the manufacturer wishes to test $H_0 : \mu = 5$ volts against $H_1 : \mu \neq 5$, using $n = 16$ units. The acceptance region is $4.88 \leq \bar{X} \leq 5.12$. Find the value of α.

(a) 0.02275

(b) 0.97725

(c) 0.95450

(d) 0.04550

(e) 0.05000

12. (Problem 4-114) Suppose that X has an exponential distribution with a mean of 10. Which of the following statements is true?

(a) $P(X < 15 \mid X > 10) = P(X > 5)$

(b) $P(X < 15 \mid X > 10) = P(X < 15)$

(c) $P(X < 15 \mid X > 10) > P(X < 5)$

(d) $P(X < 15 \mid X > 10) = P(X < 5)$

(e) $P(X < 15 \mid X > 10) < P(X < 5)$

13. (Suggested Problem 2-163) A researcher receives 20 containers of oxygen. Of those containers, 8 have oxygen that is not ionized, and the rest are ionized. Two samples are randomly selected, without replacement, from the lot. What is the probability that the second one selected is not ionized given that the first one was ionized?

(a) 0.4000

(b) 0.5789

(c) 0.4444

(d) 0.4211

(e) 0.3684

14. (Problem 11-1) An article in Concrete Research presented data on compressive strength (x) and intrinsic permeability (y) of various concrete mixes and cures. Summary quantities are $n = 12$, $\sum y_i = 457.24$, $\sum y_i^2 = 17561.7924$, $\sum x_i = 50.57$, $\sum x_i^2 = 236.4425$, and $\sum x_i y_i = 1879.9483$. Also,

$$S_{XX} = 23.33209, \quad S_{YY} = 139.42427, \quad S_{XY} = -46.93727, \quad SSR = 94.42390, \quad SSE = 45.00037.$$

Assume that the two variables are related according to the simple linear regression model. In the fitted simple linear regression model, approximately what percentage of the total variation in the response variable remains unexplained?

(a) 32.27585%

(b) 4.50004%

(c) 45.00037%

(d) 67.72400%

(e) 22.77600%

15. (Problem 12-21) Consider the R output shown in the figure below for a multiple linear regression model (15 observations, two predictors x_1 and x_2). Entries marked ?? are missing. For the test used to test whether at least one predictor is significant in the regression model, what are the degrees of freedom for numerator and denominator respectively?

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)   76.1272    10.6624   ??       ??
x1             2.6673     0.7666   3.4793   ??
x2            -2.7246     0.9865   ??       ??

Residual standard error: ?? on 12 degrees of freedom
Multiple R-squared:  ??,    Adjusted R-squared:  0.9972
F-statistic: ?? on 2 and 12 DF,  p-value: ??

Analysis of Variance
                Df    Sum Sq   Mean Sq   F value   Pr(>F)
Regression       2  1366.288   683.144     ??       ??
Residual Error  ??     ??        ??
Total           14  1369.613
```

(a) 1, 12

(b) 2, 12

(c) 14, 12

(d) 2, 14

(e) 12, 2

16. (Problem 9-26) The proportion of residents in Phoenix favoring the building of toll roads to complete the freeway system is believed to be $p = 0.4$. If a random sample of 100 residents shows that 27 or fewer favor this proposal, we will conclude that $p < 0.4$. Find the probability of committing a type II error with this procedure if $p = 0.2$.

(a) 0.95994

(b) 0.04006

(c) 0.07640

(d) 0.99598

(e) 0.22750

17. (Problem 11-1) An article in Concrete Research presented data on compressive strength (x) and intrinsic permeability (y) of various concrete mixes and cures. Summary quantities are $n = 14$, $\sum y_i = 530.24$, $\sum y_i^2 = 20338.755$, $\sum x_i = 60.16$, $\sum x_i^2 = 285.3356$, and $\sum x_i y_i = 2198.5732$. Also,

$$S_{XX} = 26.81949, \quad S_{YY} = 256.29374, \quad S_{XY} = -79.94383, \quad SSR = 238.29748, \quad SSE = 17.99627.$$

Assume that the two variables are related according to the simple linear regression model. The fitted regression line is $\hat{y} = 50.68326 - 2.98081\,x$. Which of the following is the correct interpretation of the estimated slope?

(a) For each unit increase in compressive strength, mean permeability increases by about 2.98081 units.

(b) When compressive strength is zero, mean permeability decreases by 2.98081 units.

(c) Permeability is independent of compressive strength.

(d) For each unit increase in compressive strength, mean permeability decreases by about 2.98081 units.

(e) A 2.98081-unit increase in permeability causes compressive strength to decrease by one unit.

18. (Problem 11-1) An article in Concrete Research presented data on compressive strength (x) and intrinsic permeability (y) of various concrete mixes and cures. Summary quantities are $n = 11$, $\sum y_i = 404.91$, $\sum y_i^2 = 15156.6811$, $\sum x_i = 47$, $\sum x_i^2 = 225.3286$, and $\sum x_i y_i = 1653.7781$. Also,

$$S_{XX} = 24.51042, \quad S_{YY} = 251.944, \quad S_{XY} = -76.2919, \quad SSR = 237.46857, \quad SSE = 14.47543.$$

Assume that the two variables are related according to the simple linear regression model. The fitted regression line is $\hat{y} = 50.10942 - 3.11263\,x$. Suppose that the observed value of permeability at $x = 3.00$ is 38.62. Calculate the value of the corresponding residual.

(a) -2.15153

(b) 38.62000

(c) -3.11263

(d) 2.15153

(e) 40.77153

19. (Problem 12-21) Consider the R output shown in the figure below for a multiple linear regression model (15 observations, two predictors x_1 and x_2). Entries marked ?? are missing. What is the value of the test statistic used to test whether at least one predictor is significant in the regression model?

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)   76.1272    10.6624   ??       ??
x1             2.6673     0.7666   3.4793   ??
x2            -2.7246     0.9865   ??       ??

Residual standard error: ?? on 12 degrees of freedom
Multiple R-squared:  ??,    Adjusted R-squared:  0.9972
F-statistic: ?? on 2 and 12 DF,  p-value: ??

Analysis of Variance
                Df    Sum Sq   Mean Sq   F value   Pr(>F)
Regression       2  1366.288   683.144     ??       ??
Residual Error  ??     ??        ??
Total           14  1369.613
```

(a) 3.4793

(b) 2465.4820

(c) 7.1398

(d) 683.1440

(e) 1366.2880

20. (Problem 11-1) An article in Concrete Research presented data on compressive strength (x) and intrinsic permeability (y) of various concrete mixes and cures. Summary quantities are $n = 12$, $\sum y_i = 443.99$, $\sum y_i^2 = 16679.1943$, $\sum x_i = 50.39$, $\sum x_i^2 = 235.1413$, and $\sum x_i y_i = 1788.0105$. Also,

$$S_{XX} = 23.54529, \quad S_{YY} = 251.93429, \quad S_{XY} = -76.37751, \quad SSR = 247.75755, \quad SSE = 4.17674.$$

Assume that the two variables are related according to the simple linear regression model. The fitted regression line is $\hat{y} = 50.62066 - 3.24386\,x$. What is the estimate of the error variance σ^2?

(a) 0.41767

(b) 4.17674

(c) 0.64627

(d) 0.34806

(e) 0.37970

Answer Key

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Q	Ans	Brief explanation
1	**(e)**	$z_0 = -2.29$; $p = P(Z > -2.29) = 0.98899$.
2	**(d)**	$\sigma = 20$; $P(1 < Z < 2) = 0.9772 - 0.8413 = 0.1359$.
3	**(a)**	The event is the whole sample space, so the probability is 1.0.
4	**(b)**	$\hat{\sigma}^2 = 20.57634/11$; $se = \sqrt{\hat{\sigma}^2/S_{XX}} = 0.27016$.
5	**(c)**	$\sigma_{\bar{X}} = 0.3$; $P(-1 < Z < 1) = 0.68268$.
6	**(d)**	Adding a constant shifts the mean but not the deviations, so s is unchanged.
7	**(b)**	Reject iff $\alpha \geq p = 0.012$.
8	**(c)**	$1.25 \pm 2.539(0.25/\sqrt{20}) = (1.108, 1.392)$; 1.0 is outside, so reject H_0.
9	**(b)**	$MSE = 3.325/12 = 0.27708$; $\hat{\sigma} = \sqrt{MSE} = 0.5264$.
10	**(a)**	Only $\mu = 50$ vs $\mu \neq 50$ uses a parameter with equality in H_0.
11	**(d)**	$\sigma_{\bar{X}} = 0.06$, $z = \pm 2$: $\alpha = 2(0.02275) = 0.04550$.
12	**(d)**	Memoryless: $P(X < 15 \mid X > 10) = P(X < 5) = 1 - e^{-0.5}$.
13	**(d)**	After one ionized container is removed, $\frac{8}{19} = 0.4211$ remain non-ionized.
14	**(a)**	$SSE/S_{YY} = 45.00037/139.42427 = 32.27585\%$.
15	**(b)**	Numerator $k = 2$, denominator $n - k - 1 = 12$.
16	**(b)**	$X > 27$, $X \approx N(20, 16)$: $\beta = P(Z > 1.75) = 0.04006$.
17	**(d)**	Slope -2.98081: mean permeability falls by about 2.98081 per unit of strength.
18	**(a)**	$\hat{y}(3) = 40.77153$, so $e = 38.62 - 40.77153 = -2.15153$.
19	**(b)**	$SSE = 3.325$, $MSE = 0.27708$; $F = 683.144/0.27708 = 2465.48$.
20	**(a)**	$\hat{\sigma}^2 = SSE/(n-2) = 4.17674/10 = 0.41767$.